

**[BUG] Incorrect Expectation Values of Scalar Product of Sum which is non**

<https://github.com/PennyLaneAI/pennylane/issues/3198>

ROW 26

# [BUG] Incorrect Expectation Values of Scalar Product of Sum which is non-hermitian #3198

New issue

Closed



Jaybsoni opened on Oct 21, 2022

## Expected behavior

The results of these two expectation values should be the same.

```
import pennylane as qml
from pennylane import numpy as np

dev = qml.device("default.qubit", wires=1)

@qml.qnode(dev)
def circuit(op):
    # |psi_0>: state preparation
    qml.RX(np.pi / 7, wires=0)
    qml.RY(np.pi / 3, wires=0)
    qml.RZ(np.pi / 7, wires=0)

    # Gives the expectation value relative to input operator
    return qml.expval(op)

# Define the operator sigma_x sigma_y
op = qml.prod(qml.PauliX(0), qml.adjoint( qml.PauliY(0)) )
# Define its adjoint adj(sigma_x sigma_y)
op_adj = qml.adjoint(op)
Imag = -0.5j * (op - op_adj)

assert np.allclose(qml.matrix(Imag), qml.matrix(qml.PauliZ(0)))

print(circuit(qml.PauliZ(0)))
print(circuit(Imag))
```

## Actual behavior

The results are different, a casting error occurs when computing the Eigen decomposition and the complex Eigen values of the sum operator are lost.

## Additional information

No response

## Source code

No response

## Tracebacks

No response

## System information

```
Name: PennyLane
Version: 0.27.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /opt/anaconda3/envs/pl_env/lib/python3.9/site-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, retworkx, scipy, semar
Required-by: amazon-braket-pennylane-plugin, PennyLane-Forest, PennyLane-Lightning, PennyLane-qiskit

Platform info: macOS-10.16-x86_64-i386-64bit
```

## Assignees

- AlbertMitjans
- Jaybsoni
- co9olguy

## Labels

bug

## Type

No type

## Projects

No projects

## Milestone

No milestone

## Relationships

None yet

## Development

No branches or pull requests

## Notifications

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## Participants



Visual Studio

Name: PennyLane  
Version: 0.27.0.dev0  
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.  
Home-page: <https://github.com/XanaduAI/pennylane>  
Author:  
Author-email:  
License: Apache License 2.0  
Location: /opt/anaconda3/envs/pl\_env/lib/python3.9/site-packages  
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, retworkx, scipy, semar  
Required-by: amazon-braket-pennylane-plugin, PennyLane-Forest, PennyLane-Lightning, PennyLane-qiskit

Platform info: macOS-10.16-x86\_64-i386-64bit  
Python version: 3.9.13  
Numpy version: 1.23.2  
Scipy version: 1.9.1  
Installed devices:  
- default.gaussian (PennyLane-0.27.0.dev0)  
- default.mixed (PennyLane-0.27.0.dev0)  
- default.qubit (PennyLane-0.27.0.dev0)  
- default.qubit.autograd (PennyLane-0.27.0.dev0)  
- default.qubit.jax (PennyLane-0.27.0.dev0)  
- default.qubit.tf (PennyLane-0.27.0.dev0)  
- default.qubit.torch (PennyLane-0.27.0.dev0)  
- default.qutrit (PennyLane-0.27.0.dev0)  
- braket.aws.qubit (amazon-braket-pennylane-plugin-1.7.0)  
- braket.local.qubit (amazon-braket-pennylane-plugin-1.7.0)  
- lightning.qubit (PennyLane-Lightning-0.26.0)  
- forest.numpy\_wavefunction (PennyLane-Forest-0.25.0.dev0)  
- forest.qvm (PennyLane-Forest-0.25.0.dev0)  
- forest.wavefunction (PennyLane-Forest-0.25.0.dev0)  
- qiskit.aer (PennyLane-qiskit-0.25.0.dev0)  
- qiskit.basicaer (PennyLane-qiskit-0.25.0.dev0)  
- qiskit.ibmq (PennyLane-qiskit-0.25.0.dev0)  
- qiskit.ibmq.circuit\_runner (PennyLane-qiskit-0.25.0.dev0)  
- qiskit.ibmq.sampler (PennyLane-qiskit-0.25.0.dev0)

## Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



**Jaybsoni** added **bug** on Oct 21, 2022



**Jaybsoni** assigned [co9olguy](#), [Jaybsoni](#) and [AlbertMitjans](#) on Oct 21, 2022



**Jaybsoni** mentioned this on Oct 24, 2022

[Change eig to eig to support computing the eigenvalues of non-hermitian observables #3204](#)



**Jaybsoni** closed this as **completed** on Oct 25, 2022

# Change `eih` to `eig` to support computing the eigenvalues of non-hermitian observables #3204

<> Code

Merged Jaybsoni merged 8 commits into `master` from `eigvals_bug` on Oct 25, 2022

Conversation 13

Commits 8

Checks 0

Files changed 5

+52 -7



Jaybsoni commented on Oct 24, 2022 · edited by antalszava

Contributor

## Context:

When working with linear combinations of operations. If the expressions are not fully simplified, then there may be sub-operators which are not hermitian. When asking for the eigenvalues of such operators, the default pipeline assumes hermiticity and will automatically cast to `float` types (losing information about the complex parts).

We update the pipeline to support complex eigenvalues here.

## Related Issues:

[issue 3198](#)

[\[sc-28321\]](#)



made change

6fe06ba



github-actions bot commented on Oct 24, 2022

Contributor

Hello. You may have forgotten to update the changelog!

Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



Jaybsoni added 3 commits 3 years ago

fixed bug

7223f7d

Merge branch 'master' into eigvals\_bug

Verified

548ade8

changelog

158fcfa

Jaybsoni changed the title [WIP] change `eih` to `big` [WIP] change `eih` to `eig` on Oct 24, 2022

## Reviewers

AlbertMitjans ✓  
albi3ro ✓  
co9olguy ●

## Assignees

No one assigned

## Labels

None yet

## Projects

None yet

## Milestone

No milestone

## Development

Successfully merging this pull request may close these issues.

None yet

## Notifications

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3 participants

